Problems - GOCC103: Google\u0

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Question 1

Max. score: 30.00

First subsequence

You are given the following:

- String *A* consists of lowercase English letters
- String *B* consists of lowercase English letters

You have to check whether string *A* contains string *B* as a subsequence or not. You can change at most *1* character in string *B* (except the first character).

Task

Determine the first index of occurrence of string *B* in string *A*. If not found, output *-1*.

Notes

- A string *b* is said to be a subsequence of string *a*, if *b* can be obtained from *a* by replacing at most 1 character without changing the ordering of the remaining characters.
- *1* based indexing is used.

Next Question

cases. T also denotes the number of times you have to run the *firstOccurence* function on a different set of inputs.

- For each test case:
  - The first line contains a string A consists of lowercase English letters.
  - The second line contains a string B of consists of lowercase English letters.

Output format

For each test case, print the answer in a new line.

Constraints

$1 \leq T \leq 10$
$1 \leq |A| \leq 2000$
$1 \leq |B| \leq 2000$

Code snippets (also called starter code/boilerplate code)

This question has code snippets for C, CPP, Java, and Python.

Sample input	Sample output
2	3

Therefore, you can find the first occurrence of string *B* from the 2nd index in *A*. Hence, the answer is *2*.

Function description

Complete the *firstOccurence* function provided in the editor. This function takes the following 2 parameters and returns the answer:

- *A*: Represents the first string
- *B*: Represents the second string

Input format

Note: This is the input format that you must use to provide custom input (available above the **Compile and Test** button).

- The first line contains an integer *T* denoting the number of test cases. *T* also denotes the number of times you have to run the *firstOccurence* function on a different set of inputs.
- For each test case:
  - The first line contains a string *A* consists of lowercase English letters.
  - The second line contains a string *B* of consists of lowercase

Notes

- A string *b* is said to be a subsequence of string *a*, if *b* can be obtained from *a* by replacing at most 1 character without changing the ordering of the remaining characters.
- *1* based indexing is used.

Example

Assumptions

- *A = daabe*
- *B = abe*

Approach

The subsequences of string *A* that can match with string *B* by changing at most *1* characters are:

- *A[2], A[3], A[5] : aae (requires a change of character B[2] from 'b' to 'a').*
- *A[2], A[4], A[5] : abe (requires no change).*
- *A[3], A[4], A[5] : abe (requires no change).*

- A = *lhs*
- B = *rhs*

Approach

- You can not change the first character of string B.
- Therefore, you can not find the occurrence of string B in A.

Hence, the answer is *-1*.

Note:

Your code must be able to print the sample output from the provided sample input. However, your code is run against multiple hidden test cases. Therefore, your code must pass these hidden test cases to solve the problem statement.

Limits

Time Limit: 1.0 sec(s) for each input file
Memory Limit: 256 MB
Source Limit: 1024 KB

Scoring

Score is assigned if any testcase passes

Allowed Languages

Bash, C, C++14, C++17, Clojure, C#, D, Erlang, F#, Go, Groovy, Haskell, Java 8, Java 14, Java 17, JavaScript(Node.js), Julia, Kotlin, Lisp (SBCL), Lua, Objective-C, OCaml, Octave, Pascal, Perl, PHP, Python, Python 3, Python 3.8, R(RScript), Racket, Ruby, Rust, Scala, Swift, TypeScript, Visual Basic

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The first line represents the number of test cases, *T* = *2*.

For test case 1

Given

- *A* = *abcbc*
- *B* = *cba*

Approach

The subsequences of string *A* that can match with string *B* by changing at most *1* characters are:

- *A[3], A[4], A[5] : cbc (requires a change of character B[3] from 'a' to 'c').*

Therefore, you can find the first occurrence of string *B* from the 3rd index in *A*.

Hence, the answer is *3*.

For test case 2

Given

- *A* = *lhs*
- *B* = *rhs*

Approach

- You can not change the first character of string *B*.
- Therefore, you can not find the occurrence of string *B* in *A*.

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Code snippets (also called starter code/boilerplate code)

This question has code snippets for C, CPP, Java, and Python.

Sample input

```
2
abcbc
cba
lhs
rhs
```

Sample output

```
3
-1
```

Explanation

The first line represents the number of test cases, $T = 2$.

For test case 1

Given

- A = *abcbc*
- B = *cba*

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C++17 (g++ 10.3.0)

Auto-complete ready!

```cpp
#include<bits/stdc++.h>
using namespace std;

int firstOccurence (string A, string B) {
    // Write your code here

}

int main() {

    ios::sync_with_stdio(0);
    cin.tie(0);
    int T;
    cin >> T;
    for(int t_i = 0; t_i < T; t_i++)
    {
        string A;
        cin >> A;
        string B;
        cin >> B;

        int out_;
        out_ = firstOccurence(A, B);
        cout << out_;
        cout << "\n";
    }
```

View test cases

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lish letters

tains string *B* as a
most *1* character in

of string *B* in string *A*. If

nce of string *a*, if *b* can
most 1 character without
ning characters.

Next Question >

Visible test cases   Custom input   Result

```
pdiegopavzmv
pdiegopavzmv
```

Output

```
2
2
10
13
```

Expected Correct Output

```
1
11
9
1
```

View test cases

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Question 2

Max. score: 30.00

OR XOR

You are given the following:

- Two integers *N* and *K*
- An array *A* of size *N*

Let sequence $B(B_1, B_2, \ldots, B_{2K})$ of size *2K* is a subsequence of array *A*.Let's define $F(B) = (B_1|B_2|\ldots|B_K) \oplus (B_{K+1}|B_{K+2}|\ldots|B_{2K})$.

Where $|$ denotes bitwise or operation and $\oplus$ denotes bitwise XOR operation.

Task

Determine the maximum value of *F(B)* for all possible sequences *B*.

Note: A sequence *X* is a subsequence of sequence *Y* if *X* can be obtained from *Y* by deleting several (possibly, zero or all) elements without changing the order of the remaining elements.

Example

Assumptions

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Example

Assumptions

- $N = 3$
- $K = 1$
- $A = [2, 4, 5]$

*Approach*There are 3 subsequences of length 2 which are as follows:

- $B = [2, 4]$, $F(B) = 2 \oplus 4 = 6$
- $B = [2, 5]$, $F(B) = 2 \oplus 5 = 7$
- $B = [4, 5]$, $F(B) = 4 \oplus 5 = 1$

The maximum value of $F(B)$ is 7, thus the answer is 7.

Function description

Complete the *solve* function provided in the editor. This function takes the following 3 parameters and returns a number:

- N: Represents the integer N
- K: Represents the integer K
- A: Represents an integer array of size N denoting A

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Explanation

The first line contains the number of test cases, $T = 1$.

The first test case

Given

- $N = 4, K = 2$
- $A = [1, 2, 4, 3]$

Approach

- There is *1* subsequence of length *4*.
- $B = [1, 2, 4, 3]$, $F(B) = (1$ \(|\) $2)$ \(\oplus\) $(4$ \(|\) $3) = 4$.

Thus, the answer is *4*.

Note:

Your code must be able to print the sample output from the provided sample input. However, your code is run against multiple hidden test cases. Therefore, your code must pass these hidden test cases to solve the problem statement.

Limits

Time Limit: 2.0 sec(s) for each input file
Memory Limit: 256 MB
Source Limit: 1024 KB

Scoring

Constraints

$1 \le T \le 102 \le N \le 1001 \le K \le \lfloor \frac{N}{2} \rfloor 0 \le A_i < 2^7$

Code snippets (also called starter code/boilerplate code)

This question has code snippets for C, CPP, Java, and Python.

Sample input

```
1
4
2
1 2 4 3
```

Sample output

```
4
```

Explanation

The first line contains the number of test cases, $T = 1$.

The first test case

- *K*: Represents the integer *K*
- *A*: Represents an integer array of size *N* denoting A

Input format

Note: This is the input format that you must use to provide custom input (available above the **Compile and Test** button).

- The first line contains a single integer *T* that denotes the number of test cases. *T* also denotes the number of times you have to run the *solve* function on a different set of inputs.
- For each test case:
    - The first line contains an integer *N*.
    - The second line contains an integer *K*.
    - The third line contains *N* space-separated integers denoting the array *A*.

Output format

For each test case in a new line, print the answer representing the maximum value of *F(B)* for all possible sequences *B*.

Constraints

$1 \le T \le 100$, $1 \le N \le 1000$, $1 \le K \le$ [illegible] $\le A_i \le$ [illegible]

- $N = 4, K = 2$
- $A = [1, 2, 4, 3]$

Approach

- There is *1* subsequence of length *4*.
- *B = [1, 2, 4, 3], F(B) = (1 \(|\) 2) \(\oplus\) (4 \(|\) 3) = 4.*

Thus, the answer is *4*.

Note:

Your code must be able to print the sample output from the provided sample input. However, your code is run against multiple hidden test cases. Therefore, your code must pass these hidden test cases to solve the problem statement.

Limits

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Allowed Languages

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